



DEPARTMENT OF MATHEMATICS
ACADEMIC YEAR: 2022-23 EVEN SEMESTER (BMATE201)
QUESTION BANK

MODULE-5: NUMERICAL METHODS-2

1. Solve $\frac{dy}{dx} = x^3 + y$, $y(1) = 1$ using Taylor's Series method considering up to fourth degree terms and find $y(1.1)$.
2. By using Taylor's series find y at $x=0.1$ and at $x = 0.2$ correct to 4 decimal places given

$$\frac{dy}{dx} = x^2 + y^2 \text{ with } y(0) = 1.$$

3. Use Taylor's series method to find $y(0.2)$ from $\frac{dy}{dx} = 2y + 3e^x$, $y(0) = 0$. Consider upto fourth derivative term.
4. Solve $\frac{dy}{dx} = e^x - y$, $y(0) = 2$ by using Taylor's Series method considering up to 4th degree terms and find the value of $y(1.1)$.
5. Find by Taylor's Series method the value of y at $x = 0.1$ correct to four decimal places from $\frac{dy}{dx} = x^2y - 1$, $y(0) = 1$.
6. If $y' + y + 2x = 0$, $y(0) = -1$ then find $y(0.1)$ by using Taylor's Series method considering up to 3rd degree term.
7. Using Modified Euler's Method find $y(20.2)$ given that $\frac{dy}{dx} = \log_{10}\left(\frac{x}{y}\right)$ with $y(20) = 5$ and $h = 0.2$
8. Using Modified Euler's Method find y at $x = 0.1$ given $y' = x^2 + y$ with $y(0) = 1$, $h = 0.05$
9. Given $\frac{dy}{dx} = x + \sin y$, $y(0) = 1$. Compute $y(0.4)$ with $h = 0.2$ using Euler's modified method.
10. Using Modified Euler's Method find y at $x = 0.2$ given $y' = 3x + \frac{y}{2}$ with $y(0) = 1$, taking $h = 0.2$.

11. Solve $\frac{dy}{dx} = xy$, $y(1) = 2$, find the approximate solution at $x = 1.2$ using Runge-Kutta Method.
12. Use Euler predictor and corrector formula to find y at $x=0.2$. Given $\frac{dy}{dx} = x + y$ and $y(0) = 1$ and $h = 0.2$ (correct to four decimal places)
13. Using Runge-Kutta Method of fourth order find $y(0.2)$ for the equation $y' = 3x + \frac{y}{2}$ given $y(0) = 1$ taking $h = 0.2$.
14. Using Runge-Kutta Method of fourth order find $y(0.1)$ for the equation $y' = 3e^x + 2y$ given $y(0) = 1$ taking $h = 0.1$.
15. Using Runge-Kutta Method of fourth order find $y(0.2)$ for the equation $y' = \log_{10} \left[\frac{y}{1-x} \right]$ given $y(0) = 1$ taking $h = 0.2$.
16. Using Runge-Kutta Method of fourth order find $y(0.1)$ for the equation.
$$\frac{dy}{dx} = \frac{y-x}{y+x} \quad y(0) = 1, \text{ and } h = 0.1$$
17. Apply Runge-Kutta fourth order method, to find $y(0.1)$ with $h = 0.1$ given
$$\frac{dy}{dx} + y + xy^2 = 0, y(0) = 1.$$
18. Apply Milne's predictor – corrector formulae to compute $y(0.2)$ given
$$\frac{dy}{dx} = \frac{x+y}{2} \text{ with}$$

x	0.0	0.5	1.0	1.5
Y	2.000	2.6360	3.5950	4.9680

19. Apply Milne's predictor-corrector formulae to compute $y(0.4)$ given $\frac{dy}{dx} = 2e^x y$ with

x	0	0.1	0.2	0.3
y	2.4	2.473	3.129	4.059

20. Solve $\frac{dy}{dx} = x - y^2$ with $y(0) = 0, y(0.2) = 0.002, y(0.4) = 0.0795,$



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$y(0.6) = 0.1762$. Compute y at $x = 0.8$ using Milne's Method.

21. Use Milne's method to find $y(0.4)$ from $y' = \frac{1}{2}xy$, $y(0) = 1$, $y(0.1) = 1.0025$,
 $y(0.2) = 1.0101$, $y(0.3) = 1.0228$.

20. Apply Milne's method to solve $\frac{dy}{dx} = \frac{x^2}{y^2+1}$ at $x = 1$ given $y(0) = 1$, $y(0.25) = 1.0026$, $y(0.75) = 1.0679$.

21. Given $\frac{dy}{dx} = xy + y^2$, $y(0) = 1$, $y(0.1) = 1.1169$, $y(0.2) = 1.2773$, $y(0.3) = 1.5049$ compute $y(0.4)$ using Milne's method.

22. Apply Milne's predictor-corrector formulae to compute $y(4.5)$ given that
 $5x \frac{dy}{dx} = 2 - y^2$ and

X	4.1	4.2	4.3	4.4
y	1.0049	1.0097	1.0143	1.0187